

Non-Amplifying ECG Signal Acquisition System Based on Σ - Δ A/D

Hu Liang, Zhang Yanfang, He Lesheng, Wang Weilian

(School of Information Science and Technology, Yunnan University, Kunming 650091, China)

Email:ericjackson05@qq.com

Abstract – ECG signal is a very weak signal acquired on the surface of body. In traditional ECG signal acquisition system, to get a high-SNR ECG signal, it must use all kinds of high-performance op-amp, which leads to the acquiring circuits complex and unstable. Furthermore, almost all op-amps more or less have nonlinear distortion and other disadvantages. In this paper, a non-amplifying method to acquire a high-SNR ECG signal was proposed. It is based on 24 bit Σ - Δ A/D embedded in a high performance mixture signal ARM processor-ADuC7060. The system has high SNR up to 74.8 dB, and low nonlinear distortion, original ECG can be acquired.

Keywords – Non-Amplifying ECG acquisition; high-SNR; Σ - Δ A/D; low nonlinear distortion

I. INTRODUCTION

As we know, heart disease is a biggest killer which is harmful to human's health. No matter myocardial infarction or ischemia myocardial is fatal diseases, it can takes life at any moment. Heart disease is common in our friends and neighborhoods but they didn't realized, when they work in pressure or depress in mood, which will probably lead to sudden death. Therefore, prevention of heart disease becomes more and more important in this fast-pace society. ECG (Electrocardiograph) examining is important way to diagnose and prevent heart disease in clinic.

ADC (Analog to Digital Converter) is used in traditional ECG machine. The precision of these ADC are always between 12 bit and 16 bit. ECG signal is a kind of weak signal on the surface of body. Its amplitude is just 1~4 mV level or lower. In this case, an amplifying circuit is necessary to acquire ECG signal, even ECG signal is corrupted by noise. It leads a lower SNR (Signal Noise Ratio) for acquired ECG signal. All kinds of op-amp (operational amplifier) are applied in traditional ECG machine, but it will leads to nonlinear distortion of ECG signals. Acquiring high quality ECG signal should use a new way.

In this work, a non-amplifying ECG signal acquisition system was put forward. It was based on two 24-bit Σ - Δ ADC, which embedded in an ARM chip, a high quality mixture signal processor, ADuC7060. ECG signal is sampled directly without using any op-amp. All kinds of filters, such as low-pass, high-pass, and 50Hz notch filter are implemented by using software. It makes the ECG signal non-distortion because no any op-amp

was used. A high-SNR up to 74.8 dB was got. It also reduces complexity of peripheral circuit and increasing stability of entire acquisition system.

II. PROBLEM DEFINITION

Because ECG signal is very weak, all kinds of high-precision op-amp are applied in traditional ECG machine. Although op-amp can amplify the ECG signals, there are following disadvantages when using it.

- All kinds of amplifier, especially instrumentation amplifier and isolation amplifier, have nonlinear distortions. So an original ECG signal cannot be acquire by using op-amp.

- It need a low-gain in order to prevent the output of amplifier enter the saturation, which caused by common-mode DC component, but CMR (common mode rejection) of amplifiers do not function well when it works in low-gain mode.

- Analog isolation amplifier needs dual power and works in large current, which unable supplied only by a set of battery. Therefore it is hard to manufacture portable and small-sized high precision ECG machine with isolation function.

- Sophisticated analog filters are needed to complete low-pass filter and 50Hz notch filter in front-end of isolation amplifier, otherwise noise from circuit and from environment often cause the amplifier saturated.

- Multi-leads ECG acquisition system needs multi-leads amplifier and isolation system, which make entire system in large size and high power dissipation.

In order to overcome the shortcomings above, we propose a new ECG acquisition method, a non-amplifying ECG signal acquisition system. The structured flowchart is figured in Fig.1.

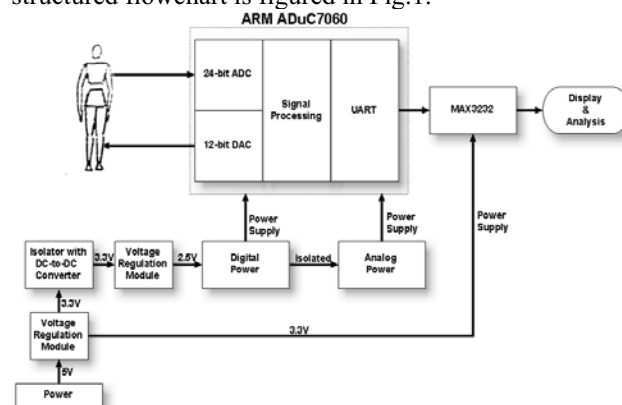


Fig.1. A flowchart of non-amplifying ECG signal acquisition system

The non-amplifying ECG signal acquisition system has these advantages:

- Not use op-amp and other various kinds of analog devices, make full use of 24-bit Σ - Δ ADC in chip to acquire data without amplification, improving SNR of ECG signal and reduce complexity of circuit.
- Using DAC in chip to provide reference voltage as right-leg driving, improving the CMR ratio, and common-mode interference will greatly reduce.
- Single power supply (battery etc.) is available in our system because we use DC-to-DC isolator, noise from power and communication module will not interfere data acquisition module, and easy to achieve miniaturization and portable.
- Baseline drift, power-frequency interference and high-frequency noise existed in ECG signal were processed by digital filters and software. Making full use of chip resources, reduce the complexity of surrounding circuit, improve stability of the system.

III. MATERIALS & METHODS

In this paper a high performance mixture signal processor, ARM ADuC7060, was used as a core of the system. It is a low power, high precision analog microcontroller which contains two high-precision Σ - Δ ADC with 24-bit. FSR (Full Scale Range) of ADC is 1200 mV in the system, so quantum step (denote Δ) of ADC is 0.0715 μ V, it can be seen in formula (1).

$$\Delta = \frac{1}{2^N} FSR = \frac{1}{2^{24}} \times 1200mV = 0.0715\mu V \dots (1)$$

Generally, the amplitude of ECG signal will not exceed 4mV, therefore non-amplifying ECG signal acquisition system is realizable in theory.

Σ - Δ ADC use a technique called oversampling which is based on the theory that a very fast, low-resolution converter can be used to take a large number of samples on an analog signal. The outputs from this oversampling process are then combined into groups, and the groups are averaged using a digital filter like an adder-accumulator. This averaging operation can increase the accuracy of the conversion as long as the input signal does not vary any faster than the sampling rate. This process also eliminates any in-band noise present in the ADC by spreading the noise across the entire sampling frequency band.

The chip ADuC7060 has high-speed operation speed because it is based on ARM7 core. Differential ADC accomplish convert analog electrical signal that from left and right wrist into digital signal, and DAC produce an analog reference voltage to restrain common-mode signal. After signal processing in chip, construct serial or wireless communication between ADuC7060 and PC to make further analysis and processing, finally, display ECG wave and analysis result on LCD or PC screen.

DC-to-DC converter is an isolator that isolated

between analog power (ground) and digital power (ground), it is a very important and necessary device. It's almost impossible to accomplish single power supply without DC-to-DC converter. It ensures noise in analog devices not to have influence on digital ones, ECG signal will covered by noise without it. Power supply module is briefly shown in Fig.2.

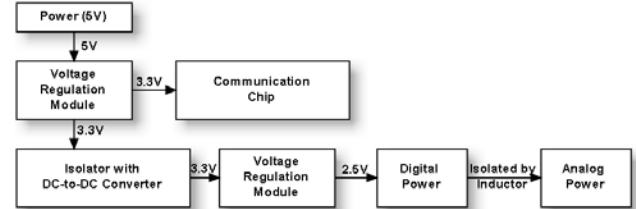


Fig.2. Power supply module diagram

The serial voltage level switch chip max3232 compose the communication module. A brief chart of communication module was illustrated in Fig.3.

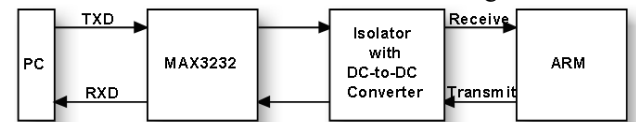


Fig.3. Communication module diagram

A flow chart of program in ADuC7060 is shown in Fig.4. As we can see from this flow chart, ADC, digital filter and communication are three important parts of the program.

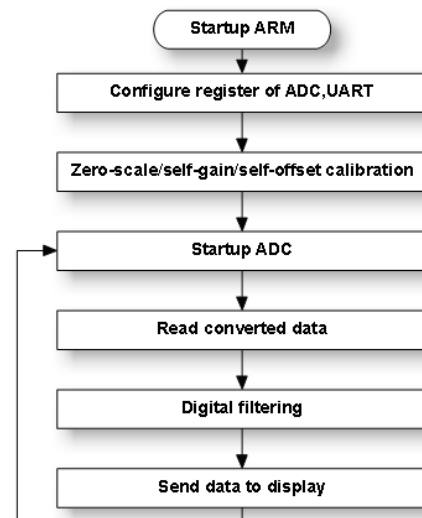


Fig.4. ARM program flow chart

There are three digital filters implemented in ADuC7060 chip, they are 50Hz notch filter, 0.05Hz high-pass filter, and 160Hz low-pass filter. In this system, any DSP chip was not used, because of the following reasons.

- Using DC-to-DC isolator and noise-restrain circuit layout method, so original signal after ADC has little noise. Two-order filter has competence for filter ECG signal and it reduces computation greatly.
- Less chips and simple peripheral circuit could help to extend lifetime of battery.

• As DSP chip is expensive, filtering in ARM could reducing production cost.

IV. RESULTS

A. Waveform test

A comparison of traditional ECG signal acquisition method and non-amplifying ECG signal acquisition method can be seen in Fig.5.

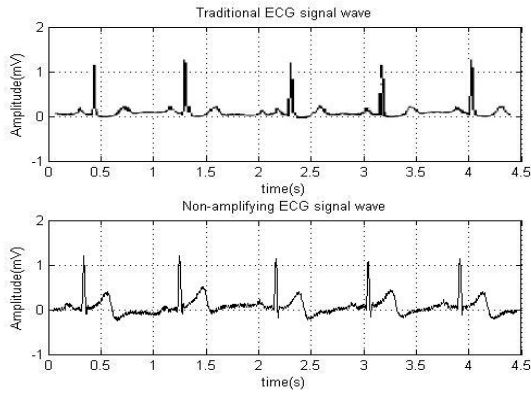


Fig.5. Traditional ECG and non-amplifying ECG signal wave

non-amplifying ECG signal wave is shown in under part of Fig.5, P wave, QRS wave and T wave are easy to distinguish, and baseline drift, industrial-frequency noise and high-frequency noise are not exist. Non-amplifying ECG signal wave is even better compared with traditional ECG wave.

B. digital filter test

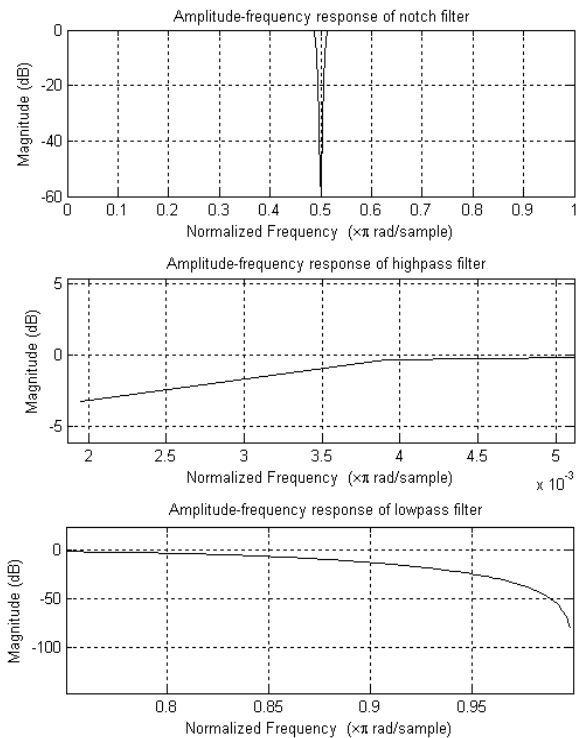


Fig.6. Amplitude-frequency response of filters

The amplitude-frequency response of notch filter, high-pass filter and low-pass filter is shown in Fig.6.

ECG wave without filter is shown in upper part of Fig.7. Industrial-frequency noise is obviously, P wave is covered by noise. The effect after filtering is shown in under part of Fig.7, little noise exists after filtering.

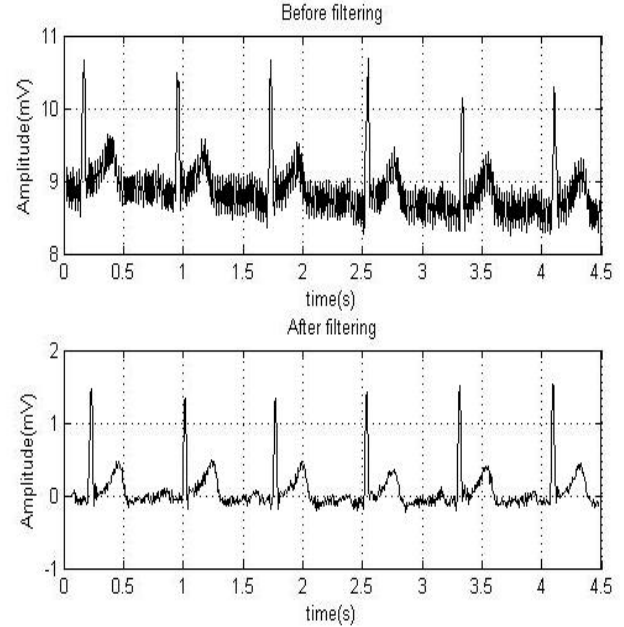


Fig.7. Wave before filtering and after filtering

C. SNR test

SNR is a measurement used in science and engineering to quantify how much a signal has been corrupted by noise. The SNR is defined as formula (2).

$$SNR_{dB} = 10 \log \left(\frac{P_{signal}}{P_{noise}} \right) = 10 \log \left(\frac{A_{signal}^2}{A_{noise}^2} \right) = 20 \log \left(\frac{A_{signal}}{A_{noise}} \right) \dots (2)$$

Where P and A are average power and RMS amplitude of voltage.

The RMS (root mean square) voltage of ECG signal and noise is hard to acquire, because there is no standard ECG, different people has different ECG. In order to calculate SNR of non-amplifying ECG signal acquisition system approximately, a single frequency sine signal produced by a signal generator is sampled by the system. Then produce a standard sine signal of same frequency in MATLAB. The energy of sampled signal is maximum self-correlation (denote r_{xx}) of it, which equals 1 after normalization, and the energy of standard sine signal is maximum cross-correlation (denote r_{xy}) of standard signal and sampled signal, which value less than 1 after normalization. Then subtract r_{xy} from r_{xx} is the energy of noise. Formula (3) is used to calculate SNR of system.

$$SNR_{dB} = 10 \log \left(\frac{P_{signal}}{P_{noise}} \right) = 10 \log \left(\frac{r_{xy}}{r_{xx} - r_{xy}} \right) = 74.8423 dB \dots (3)$$

V. CONCLUSION & DISCUSSION

Non-amplifying ECG signal acquisition has high SNR and low nonlinear distortion. The original ECG

signals can be acquired by this way. It is useful to do further research in ECG and obtain valuable diagnosis information in clinic. There are only few components on circuit board except for ADuC7060 chip and DC-to-DC converter, improving stability of system, reducing power dissipation and small in size. The system is easy to portable and fit for family use.

Not only ADuC7060 chip, but also any chip with high-precision and low-noise Σ - Δ ADC can construct a non-amplifying ECG signal acquisition system. And any type of high quality isolator with DC-to-DC converter chip also available.

This paper is supported by Research Programs of Science and Technology Commission Foundation of Yunnan Province (No.2008CD080).

ACKNOWLEDGMENT

Thanks for my teacher He Lesheng and Wang Weilian, who provided technical support and help me a lot in my study and daily life, He Lesheng is the corresponding author of this paper. Thanks to committee of ICEMI who give me an opportunity to bring forth my research.

REFERENCES

- [1] FRANCO S. Design with operational amplifiers and analog integrated circuit[M]. Xi'an: Xi'an Jiaotong University Press, 2009.
- [2] PENG J. Operational amplifier and its application[M]. Beijing: Science Press, 2008.
- [3] LI G, WANG Y, ZENG R L, et al. High-performance mixed signals ARM-ADuC7XXX principle and application development[M]. Beijing: Publishing House of Electronics Industry, 2009.
- [4] CHEN S S, LEI J, GUO Q. Electronic measurement and instrumentation[M]. Beijing: Publishing House of Electronics Industry, 2007.
- [5] GUO J R. Electrocardiography[M]. Beijing: People's Medical Publishing House, 2005.
- [6] SU J M, RUAN S Y. A practical course of MATLAB[M]. Beijing: Publishing House of Electronics Industry, 2005.

AUTHOR BIOGRAPHY

Hu Liang was born in Shijiazhuang, China, in 1988. He received B.Sc. from Yunnan University, China, in 2010. Now he is a M.Sc. candidate in Yunnan University, China. His research interests include biomedical signal processing, electronic measurement and instruments.